

UNIT-3

Context-Free Grammar

$$G = (V, T, P, S)$$

V : set of variables | non-terminals (upper-case letters)

T : set of terminals | (lower-case letters)

P : set of productions | Rules

$$V \rightarrow (VUT)^*$$

S : starting variable

Ex:

1) $S \rightarrow AB$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

2) $S \rightarrow aAb$

$$A \rightarrow aA \mid bB \mid \epsilon$$

3) $S \rightarrow AB$
 $aA \rightarrow \epsilon$

not CFG as it starts with lower case 'a'

CFG Construction

1) CFG from Finite Automata

1) Obtain grammar to generate string with any number of a's



$$S(s, a) \rightarrow s$$

CFG

$$S \rightarrow aS \quad (s \rightarrow \text{input symbol, state})$$

$$S \rightarrow \epsilon$$

derivation

$$S \Rightarrow aS$$

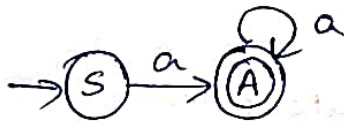
$$\Rightarrow aas$$

$$\Rightarrow aaas$$

$$S \Rightarrow aaaS =$$

$$S \Rightarrow aaaa$$

2) Obtain CFG to generate string consisting of atleast one 'a'



$$S(S, a) \rightarrow A$$

$$S(A, a) \rightarrow A$$

CFG

$$S \rightarrow aA$$

$$A \rightarrow aA$$

$$A \rightarrow \epsilon$$

can be written as $S \rightarrow aA$ $A \rightarrow aA / \epsilon$

derivation

check for aaaa

$$S \Rightarrow aA$$

$$\Rightarrow aaA$$

$$\Rightarrow aaaa$$

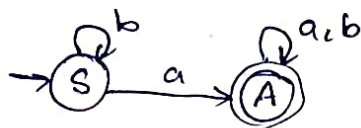
$$\Rightarrow aaaaa$$

$$\Rightarrow aaaaa\epsilon$$

$$S \Rightarrow aaaa$$

3) string consisting of any number of a's and b's

4) Strings of a's and b's consisting of at least one 'a'



$S \Rightarrow b^*$

$$\delta(S, b) = S$$

$$\delta(S, a) = A$$

$$\delta(A, a) = A$$

$$\delta(A, b) = A$$

CFG

$$S \rightarrow bs | aA$$

$$A \rightarrow aA | bA | \epsilon$$

II CFG from RE

1) strings of a's and b's ending with 'ab'

$$(a + b)^* ab$$

sl: replace repeatable terms with variables

$$(a + b)^* \rightarrow A$$

$$S \rightarrow Aab$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

aaabab is valid

$$S \Rightarrow Aab$$

$$S \Rightarrow bAab$$

$$S \Rightarrow aAab$$

$$S \Rightarrow aaAab$$

$$S \Rightarrow aabAab$$

$$S \Rightarrow aabab$$

2) strings of a's and b's having substring 'ab'

$$(a + b)^* ab (a + b)^*$$

$$\downarrow$$

$$A$$

$$\downarrow$$

$$A$$

$$S \rightarrow AabA$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

III CFG for non regular languages

1) $L = \{a^n b^n : n \geq 0\}$

$$S \rightarrow asb \mid \epsilon$$

to get aaabbb

$$S \Rightarrow asb$$

$$S \Rightarrow aasbb$$

$$S \Rightarrow aaasbbb$$

$$S \Rightarrow aaa bbb$$

$$2) L = \{a^{n+1}b^n : n \geq 0\}$$

$$S \rightarrow asb | a$$

$$3) L = \{a^n b^{n+1} : n \geq 0\}$$

$$S \rightarrow asb | b$$

$$4) L = \{a^n b^{2n} : n \geq 0\}$$

$$S \rightarrow asbb | \epsilon$$

$$5) L = \{a^n b^{2n} : n \geq 1\}$$

$$S \rightarrow asbb | \underline{abb}$$

min. requirement

$$6) \underbrace{0^m}_A \underbrace{1^m}_B \underbrace{2^n}_C \quad m \geq 1, n \geq 0$$

$$S \rightarrow \overbrace{AB}$$

$$B \rightarrow 2B | 2 \in$$

$$S \rightarrow AB$$

$$A \rightarrow 0A | 0 \in$$

$$B \rightarrow 2B | \epsilon$$

$$A \rightarrow 0A | 0 \in$$

$$7) ww^R, w \in (a+b)^*$$

$$L = \{abbbbba, baaaab, abaaba\}$$

$$S \rightarrow aSa | bSb | \epsilon$$

$$\underline{baaaab}$$

$$S \Rightarrow bSb$$

$$\Rightarrow baSab$$

$$\Rightarrow baaSaab \quad | S \Rightarrow \epsilon$$

$$\Rightarrow baaaab$$

$$\Rightarrow a b S b a$$

$$\Rightarrow abbsbba$$

$$\Rightarrow abbbbba$$

$$8) w \in w^R, w \in (a+b)^*$$

$$L = \{ aacaa, bacab \}$$

$$S \rightarrow aSa | bSb | c$$

$$9) w : n_a(w) = n_b(w), w \in (a+b)^*$$

$$S \rightarrow aSb | bSa | \epsilon | SS | \epsilon$$

ababba

$$S \Rightarrow aSb$$

$$aSbSa$$

$$aSaSa$$

$$aba$$

$$SS$$

$$aSbS$$

$$abS$$

$$abSS$$

$$ab aSbSa$$

$$ababba$$

$$10) w : n_a(w) = n_b(w) + 1$$

$$S \rightarrow aSb | bSa | SS | a$$

11) Generate a string of balanced parenthesis

$$S \rightarrow (S) | \{S\} | [S] | \epsilon | SS | \epsilon () | \epsilon [] | \epsilon \{ \}$$

min requirement

$$(() [] \{ () \})$$

$$S \Rightarrow SS (S)$$

$$\Rightarrow SSS (SS)$$

$$\Rightarrow (SSS)$$

$$\Rightarrow (() [] \{ () \})$$

$$\Rightarrow (() [] \{ () \})$$

$$12) 0^i 1^j, i \neq j, i, j \geq 0$$

$$S \rightarrow \cancel{00S1} \mid \cancel{00S1} \mid \epsilon$$

$$S \rightarrow \cancel{0S1} \mid \cancel{0S1} \mid \epsilon$$

$$S \rightarrow 0S1 \mid A \mid B$$

$$A \rightarrow 0A \mid 0$$

$$B \rightarrow 1B \mid 1$$

$$000011$$

$$S \rightarrow 0S1$$

$$\rightarrow 00S11$$

$$\rightarrow 000A11$$

$$\rightarrow 000011$$

$$13) a^{n+2} b^m, n \geq 0, m > n$$

$$L = \{ aabbb, aaabbbbb \}$$

$$S \rightarrow aaSbbb$$

$$i) n=0, m \geq 1$$

$$\underline{aabb}^*$$

$$ii) n=1, m \geq 2$$

$$aaabbb^*$$

$$\cancel{S \rightarrow aaaSb}$$

$$S \rightarrow aAB$$

$$A \rightarrow aAb \mid \epsilon ab$$

$$B \rightarrow bB \mid \epsilon$$

14) $a^n b^m, n \geq 0, m \geq n$

i) $n=0, m \geq 1$

bb^*

ii) $n=1, m \geq 2$

$abbb^*$

iii) $n=2, m \geq 3$

$aabbbb^*$

$S \rightarrow AbB$

$A \rightarrow aAb | \epsilon$

$B \rightarrow bB | \epsilon$

15) $a^n b^{n-3}, n \geq 3$

i) when $n=3$

aaa

ii) $n=4$

$aaaab$

iii) $n=5$

$aaaaabb$

$S \rightarrow aaaA$

$A \rightarrow aAb | \epsilon$

16) $L = L_1 L_2$

$L_1 = \{a^n b^n, n \geq 0\}$

$L_2 = \{a^n b^{2n}, n \geq 0\}$

$S \rightarrow S_1 S_2$

$S_1 \rightarrow aS_1b | \epsilon$

$S_2 \rightarrow aS_2bb | \epsilon$

17) Strings of 0's and 1's having substring 000

$$S \rightarrow A000A$$

$$A \rightarrow 0A|1A|\epsilon$$

18) $a^n b^m c^k$ $n+2m=k$, $n, m \geq 0$

$$a^n b^m c^{n+2m}$$

$$a^n b^m c^n c^{2m}$$

$$a^n b^m c^{2m} c^n$$

$$S \rightarrow aAc|\epsilon$$

$$A \rightarrow a b A c c | \epsilon$$

$$B \rightarrow b B c c | \epsilon$$

5/12

Derivations / Parsing

- Left-most derivation (LMD)
- Right-most derivation (RMD)

Eg: $S \rightarrow AB$

$$A \rightarrow aA|\epsilon$$

$$B \rightarrow bB|\epsilon$$

LMD

$$S \Rightarrow AB$$

$$\Rightarrow aAB \quad // A \rightarrow aA$$

$$\Rightarrow aaAB \quad // A \rightarrow aA$$

$$\Rightarrow aaB \quad // A \rightarrow \epsilon$$

$$\Rightarrow aabB \quad // B \rightarrow bB$$

$$\Rightarrow aabbb \quad // B \rightarrow bB$$

$$\Rightarrow aabbbb \quad // B \rightarrow bB$$

$$\Rightarrow aabbbb \quad // B \rightarrow \epsilon$$

RMD

$$S \Rightarrow AB$$

$$\Rightarrow AbB \quad // B \rightarrow bB$$

$$\Rightarrow Abbb \quad // B \rightarrow bB$$

$$\Rightarrow Abbbb \quad // B \rightarrow bB$$

$$\Rightarrow Abbbb \quad // B \rightarrow \epsilon$$

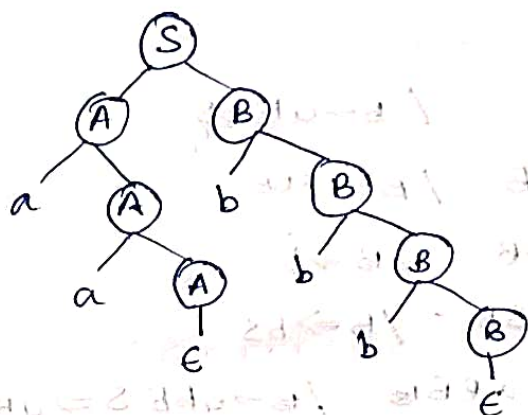
$$\Rightarrow aAbbbb \quad // A \rightarrow aA$$

$$\Rightarrow aaAbbbb \quad // A \rightarrow aA$$

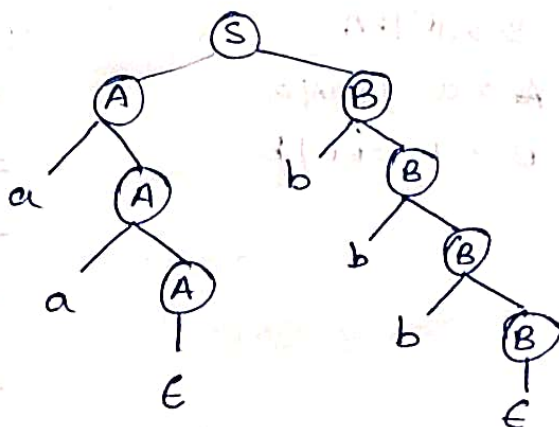
$$\Rightarrow aabbbb \quad // A \rightarrow \epsilon$$

Derivation Tree (Parse Tree)

LMD



RMD



obtain the left most derivation for the string aaabbb-aab

LMD
 $S \rightarrow AB | aab$
 $A \rightarrow a | Aa$
 $B \rightarrow b$

$S \Rightarrow aab$

$S \Rightarrow AB$

$\Rightarrow AaB \parallel A \rightarrow Aa$

$\Rightarrow aaB \parallel A \rightarrow a$

$\Rightarrow aab \parallel B \rightarrow b$

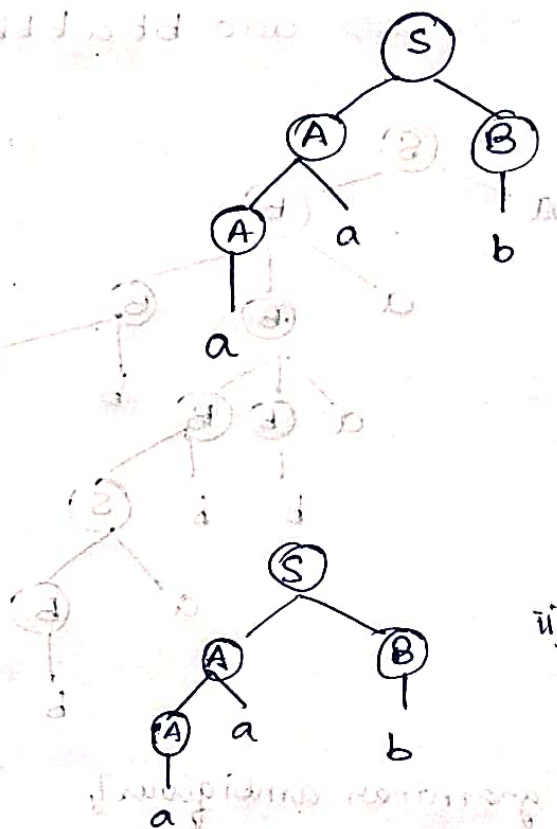
RMD

$S \Rightarrow AB$

$\Rightarrow Ab \parallel B \rightarrow b$

$\Rightarrow Aab \parallel A \rightarrow Aa$

$\Rightarrow aa.b \parallel A \rightarrow a$



ii) $S \Rightarrow aab$

$S \Rightarrow aab$

↓

can be LMD or RMD

Ambiguous grammar

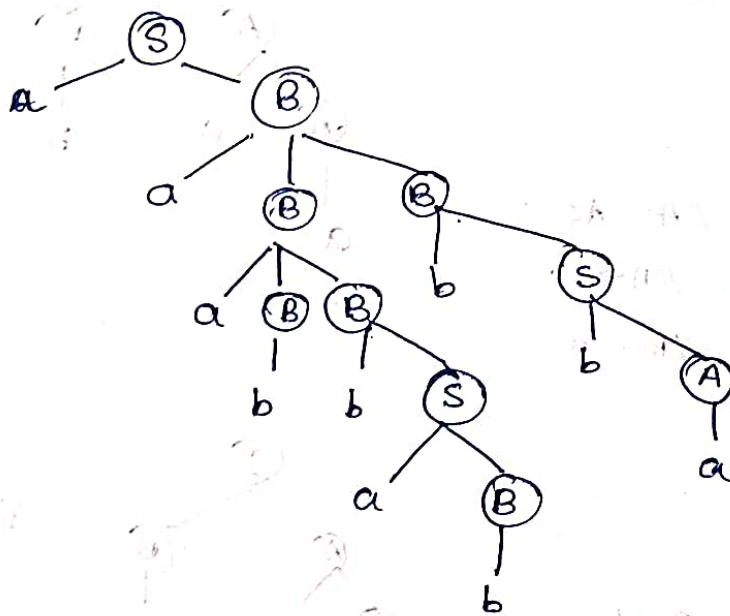
1) 2 LMDs or 2) 2 RMDs or 3) 1 LMD & 1 RMD

if structure of PT is same $\Rightarrow$ ambiguous

1) Obtain the RMD for the question $aaabbaabba$

$S \rightarrow aB | bA$
 $A \rightarrow aS | bA | a$
 $B \rightarrow bS | aBB | b$

$S \Rightarrow aB$
 $\Rightarrow aaBBB \quad / B \Rightarrow aBBB$
 $\Rightarrow aaaBBB \quad / B \Rightarrow aBB$
 $\Rightarrow aaabBB \quad / B \Rightarrow b$
 $\Rightarrow aaabbsB \quad / B \Rightarrow bs$
 $\Rightarrow aaabbaBBB \quad / B \Rightarrow aBB \quad S \Rightarrow aB$
 $\Rightarrow aaabbabB \quad / B \Rightarrow b$
 $\Rightarrow aaabbabbs \quad / B \Rightarrow bs$
 $\Rightarrow aaabbabbsA \quad / B \Rightarrow S \Rightarrow bA$
 $\Rightarrow aaabbabbsa \quad / A \Rightarrow a$



2) Is the following grammar ambiguous?

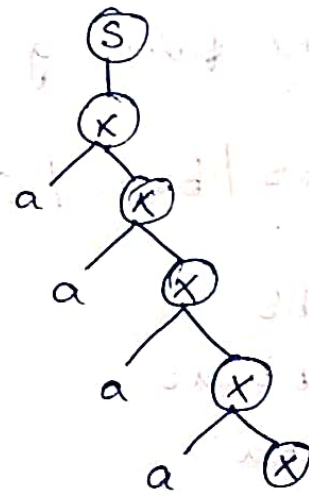
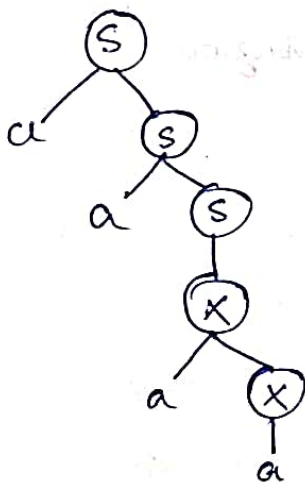
$S \rightarrow aS | X$
 $X \rightarrow aX | a$

$S \Rightarrow aS$
 $\Rightarrow aas \quad // S \Rightarrow as$
 $\Rightarrow aax \quad // S \Rightarrow X$
 $\Rightarrow aaax \quad // X \Rightarrow ax$
 $\Rightarrow aaaa \quad // X \Rightarrow a$

$S \Rightarrow aX$
 $\Rightarrow ax \quad // X \Rightarrow ax$
 $\Rightarrow aax \quad // X \Rightarrow ax$
 $\Rightarrow aaax \quad // X \Rightarrow ax$
 $\Rightarrow aaaa \quad // X \Rightarrow a$

RMD

(LMD/RMD)



The graphs are different, $\therefore$ ambiguous grammar

3) Show that the following grammar is ambiguous

I/p string : $ibtibtaea$

$S \rightarrow iCtS / iCtSeS / a$

$C \rightarrow b$

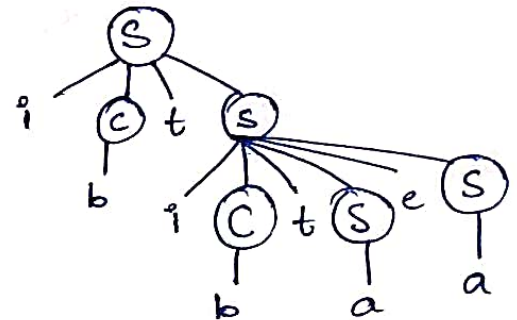
RMD

$S \Rightarrow iCtS$

$\Rightarrow ibtS // C \rightarrow b$

$\Rightarrow ibtiCtSeS // S \rightarrow iCtSeS$

$\Rightarrow ibtibtaea // C \rightarrow b$
 $S \Rightarrow a$

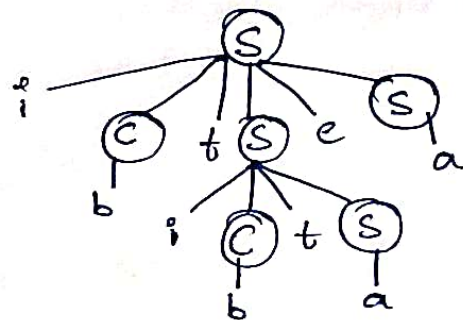


R.

$S \Rightarrow iCtSeS$

$\Rightarrow iCtiCtSeS // S \rightarrow iCtS$

$\Rightarrow ibtibtaea // C \rightarrow b$
 $S \Rightarrow a$



The graphs are different, $\therefore$ ambiguous grammar

4) Show that the following grammar is ambiguous

$$S \rightarrow aSbS \mid bSaS \mid \epsilon$$

RMD

$$S \Rightarrow aSbS$$

$$\Rightarrow aSbbSaS$$

$$\Rightarrow abba$$

LMD

$$S \Rightarrow bSaS$$

$$bSaSbS$$

$$abSaSbS$$



The above are different derivations for the string 'abba'.

III Eliminating useless symbols

- Eliminating non-generating symbols
- Eliminating non-reachable symbols

Reachable symbols

- S is reachable
- $A \rightarrow \alpha$, if A is reachable, then all the symbols in α are reachable

Non Generating symbols

$GS \equiv \{a\}$

- All terminating symbols
- $A \rightarrow \alpha$, all the symbols of α are in GS list, then A is also in GS

1) $S \rightarrow AB | AC$

$A \rightarrow aA | bAa | a$

$B \rightarrow bbA | aB | AB$

$C \rightarrow aCa | aD$

$D \rightarrow aD | bC$

$GS = \{a, b, A, B, S\}$

$NGS = \{C, D\}$

Remove all C and D productions

$S \rightarrow AB | a$

$A \rightarrow aA | bAa | a \Rightarrow$ free from NGS

$B \rightarrow bbA | aB | AB$

2) ~~$S \rightarrow AB$~~ $S \rightarrow a | bB | aA'$

~~$A \rightarrow a$~~

Eliminate

NRS

$A \rightarrow aB$

$B \rightarrow a | Aa$

$D \rightarrow ddd$

$RS = \{a, S, a, B, b, A\}$

$NRS = \{D, d\}$

eliminate $D \rightarrow ddd$

CNF - Chomski Normal Form

Grammar should be of form

$$A \rightarrow BC$$

$$A \rightarrow a$$

$$S \rightarrow a | Bb | aA$$

$$A \rightarrow aB$$

$$B \rightarrow a | Aa$$

$$S \rightarrow a | BX | YA$$

$$A \rightarrow YB$$

$$B \rightarrow a | AY$$

$$X \rightarrow b$$

$$Y \rightarrow a$$

1) Eliminate unit productions, ϵ -productions, and useless ~~pro~~ symbols
put the resulting grammar in CNF

$$S \rightarrow ABC' | BaB$$

$$A \rightarrow aA | BaG | aaa$$

$$B \rightarrow bBb | a | D$$

$$G \rightarrow GA | AG$$

$$D \rightarrow \epsilon$$

a) eliminate ϵ -prods

rule) i) RHS is ϵ

$\Rightarrow$

ii) ϵ -list is in RHS

after NS list, substitute the symbols with ϵ

NS = $\{D, B\}$ - ϵ substitution

$$S \rightarrow ABC' | AG | BaB | aB | BaA$$

$$A \rightarrow aA | BaG | aG | aaa$$

$$B \rightarrow bBb | bb | a | D$$

$$G \rightarrow GA | AG$$

$$D \Rightarrow \epsilon$$

b) eliminate unit productions

... $A \rightarrow X$ where both A and X are variables

UP	Prods (Non Unit)
(S, S)	$S \rightarrow ABC \mid AC \mid BaB \mid aB \mid Ba \mid a$
(A, A)	$A \rightarrow aA \mid BaC \mid aC \mid aaa$
(B, B)	$B \rightarrow bBb \mid bb \mid a$
(C, C)	$C \rightarrow CA \mid AC$
(D, D)	D
(B, D)	$B \rightarrow \dots$ (D does not exist)

Useless symbols $\begin{cases} \text{NGS} \\ \text{NRS} \end{cases}$

$$GS = \{a, b, B, S, A\}$$

$$NGS = \{C\}$$

Remove all C productions

$$S \rightarrow BaB \mid aB \mid Ba \mid a$$

$$A \rightarrow aA \mid aaa$$

$$B \rightarrow bBb \mid bb \mid a$$

$$RS = \{S, a, B, b\}$$

$$S \rightarrow BaB \mid aB \mid Ba \mid a$$

$$NRS = \{A\}$$

$$B \rightarrow bBb \mid bb \mid a$$

remove all A productions

CNF

$$S \rightarrow BX \mid YB \mid aBY \mid a$$

$$B \rightarrow ZP \mid PP \mid a$$

$$X \rightarrow \cancel{a}B$$

$$Y \rightarrow a$$

$$Z \rightarrow BP$$

$$P \rightarrow b$$

2)

$$S \rightarrow AAA \mid B$$

$$A \rightarrow aA \mid B$$

$$B \rightarrow \epsilon$$

a) eliminate ϵ - productions

$$NS = \{B, A, S\}$$

$$S \rightarrow$$

II) Greibach Normal Form (GNF)

$$V \rightarrow TV^*$$

$$A \rightarrow BC \quad \times$$

$$B \rightarrow bB/b \quad \checkmark$$

$$C \rightarrow cC/c \quad \checkmark$$

not GNF

$$A \rightarrow aAB/a \quad \checkmark$$

$$B \rightarrow bB/b \quad \checkmark$$

GNF

Convert the following grammar to GNF

$$A \rightarrow BC$$

$$B \rightarrow CA/b$$

$$C \rightarrow AB/a$$

steps

i) should be ⁿCNF

ii) Rename variables as $A_1, A_2, \dots, A_n$

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1/b$$

$$A_3 \rightarrow A_1 A_2/a \quad \times$$

iii) Consider the production where $i > j$

$$A_i \rightarrow A_j \alpha$$

$$A_3 \rightarrow \boxed{A_1} A_2/a$$

Substitute A_1 productions

$$A_3 \rightarrow \boxed{A_2} A_3 A_2/a$$

$$A_3 \rightarrow \boxed{A_3 A_1} A_3 A_2/b, \boxed{A_3} A_3 A_2/a$$

- eliminate ~~extra~~ coz its left recursion

Note: Left recursion!

$$A \rightarrow A\alpha_1 \mid A\alpha_2 \mid \dots \mid \beta_1 \mid \beta_2$$

$\equiv$

$$A \rightarrow \beta_1 z \mid \beta_2 z \mid \dots \mid \beta_1 \mid \beta_2$$

$$z \rightarrow \alpha_1 z \mid \alpha_2 z \mid \dots \mid \alpha_1 \mid \alpha_2$$

using this

iv)

$$A_3 \rightarrow A_3 \overbrace{A_1 A_2 A_2}^{\alpha_1} \mid b \overbrace{A_3 A_2}^{\beta_1} \mid a \quad \beta_2$$

$$A_3 \rightarrow b A_3 A_2 z \mid a z \mid b A_3 A_2 \mid a$$

- GNF ✓

$$z \rightarrow A_1 A_3 A_2 z \mid A_1 A_3 A_2$$

v) substitute A_3 prods

$$A_2 \rightarrow A_3 A_1 \mid b$$

$$A_2 \rightarrow \boxed{b A_3 A_2 z} A_1 \mid \boxed{a z} A_1 \mid \boxed{b A_3 A_2} A_1 \mid \boxed{a} A_1 \mid b$$

$$A_1 \rightarrow A_2 A_3$$

$$A_1 \rightarrow \boxed{b A_3 A_2 z} A_1 A_3 \mid \boxed{a z} A_1 A_3 \mid \boxed{b A_3 A_2 A_1} A_3 \mid \boxed{a A_1} A_3 \mid \boxed{b A_3}$$

$$z \rightarrow A_1 A_3 A_2 z \mid A_1 A_3 A_2$$

$$z \rightarrow \boxed{b A_3 A_2 z A_1 A_3} A_3 A_2 z \mid \boxed{a z A_1 A_3 A_3 A_2 z} \mid b A_3 A_2 A_1 A_3$$

$$A_3 A_2 z \mid \boxed{a A_1 A_3} A_3 A_2 z \mid \boxed{b A_3} A_3 A_2 z \mid b A_3 A_2 z A_3$$

$$A_3 A_2 \mid \boxed{a z A_1 A_3} A_3 A_2 \mid \boxed{b A_3 A_2 A_1 A_3} A_3 A_2 \mid \boxed{a A_1 A_3} A_3 A_2 \mid \boxed{b A_3 A_3 A_2}$$

$$2) \quad S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

i) CNF

ii) Rename

$$A_1 \rightarrow A_2 A_2 \mid 0$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

$$A_i \rightarrow A_j \quad i > j$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

substitute A_1 periods

$$A_2 \rightarrow A_2 \boxed{A_2 A_1} \mid 0 A_1 \mid 1$$

apply left recursion

$$A_2 \rightarrow 0 A_1 Z \mid 1 Z \mid 0 A_1 \mid 1$$

$$Z \rightarrow A_2 A_1 Z \mid A_2 A_1$$

Substitute A_2 periods

$$A_1 \rightarrow A_2 A_2 \mid 0$$

$$A_1 \rightarrow 0 A_1 Z A_2 \mid 1 Z A_2 \mid 0 A_1 A_2 \mid 1 A_2 \mid 0$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

$$A_2 \rightarrow 0 A_1 Z A_2 A_1 \mid 1 Z A_2 A_1 \mid 0 A_1 A_2 A_1 \mid 1 A_2 A_1 \mid 0 A_1 \mid 1$$

$$Z \rightarrow 0 A_1 Z A_2 A_1 A_1 Z \mid 1 Z A_2 A_1 A_1 Z \mid 0 A_1 A_2 A_1 A_1 Z \mid 1 A_2 A_1 A_1 Z \mid$$

$$0 A_1 A_1 Z \mid 1 A_1 A_1 Z \mid 0 A_1 Z A_2 A_1 A_1 \mid 1 Z A_2 A_1 A_1 \mid 0 A_1 A_2 A_1 A_1 \mid 1 A_2 A_1 A_1 \mid 0 A_1 A_1 \mid 1 A_1$$